Game Production Project Management Kickoff

MTD.ba/DA.ba SS 2026, Hagenberg

Project Management: Scrum for Games

Scrum

- → Tasks & Work packages
  - → Tasks, Stories & Epics
  - → Task status (Backlog, To Do, In Progress, ...)
- → Cost estimate
  - → Story Points
- → Work phases
  - → Sprints
- → Communication
  - → Daily Scrum, Sprint Review, ...
- → Roles
  - → Project Owner, Scrum Master, Development Team
- → Evaluation & Control
  - → Burndown Charts, ...

Tasks & Work Packages

The whole project is broken down into Stories & Tasks.

(User) Story

A feature of your game that can be implemented by the responsible person in < 1 sprint.

Task

Work packages belonging to a story.

Locomotion Programming

Locomotion (Run, Jump)

Specials (Wall Jump, Double Jump, Dash,...)

Fight Moves Programming

Fighting

Fighting Specials (Sword Jump,...)

Cost Estimate – How?

“Ideal“ vs. “Maximum“

If all goes well

If something goes wrong

Developers estimate themselves

Locomotion programming:

4–12h

Locomotion Programmierung

Locomotion (Run, Jump)

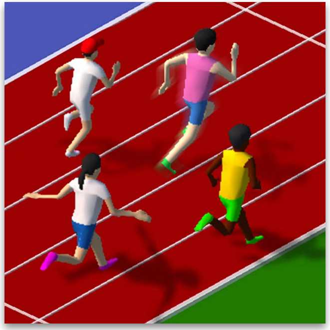
Specials (Wall Jump, Double Jump, Dash,...)

Fight Moves Programmierung

Fighting

Fighting Specials (Sword Jump,...)

Work Phases & Sprints



The project work is divided into **sprints**.

→ 1–4-weeks work phase.
→ Includes a predefined set of stories (for all team members).
→ Usually has a theme ("Controls & art prototype")
→ In the end, there is (ideally) a playable intermediate result.

After a sprint, a review meeting is held to discuss suggestions for improvement in the planning and process of the next sprint.

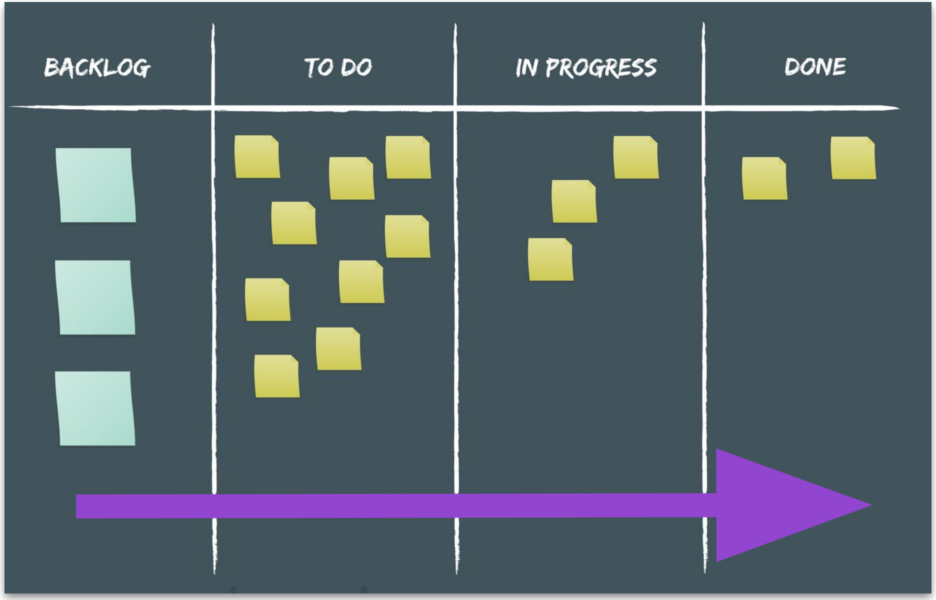
Status & Scrum Board

Stories have a **status**.

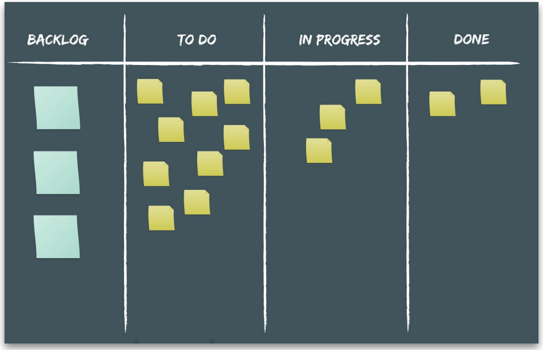
→ Updated by the responsible developers.

→ Often represented as the position of a story card on a Scrum board.

→ The more progress, the further the card is located to the right.



Typical Status Columns



Backlog/No Status

This is where all stories go at the beginning.
On GitHub, it's not there at the beginning but it will appear whenever there are items that haven't been assigned a status yet.

Todo:

Contains all stories that belong to the current sprint. Developers take their next stories (and tasks) from here.

In Progress

Everything that is being worked on right now.
Advanced boards often have an "In Review" column for stories that have been implemented but are being reviewed for quality control.

Done

Everything that is done.

Also popular/possible:

One Ready/backlog column **per sprint** for planning ahead.

Daily Scrum / Standup

During sprints, all developers meet in the morning for 10 min and discuss while standing:

→ What did I do yesterday?
→ What am I doing today?
→ Are there any dependencies?

Why standing?

→ *So that it's uncomfortable and the meeting is kept short!* 😬

Prepare for Production: What to Do

Your Project Management Preparations

1. Access your project board on GitHub.
   → We'll do this together.

2. Break down your entire project into stories.
   → Create the stories as GitHub issues in your repository.
   → Assign a person responsible for the story.
   → Tasks can be created later.
   → Resource: Game_Production_Project_Planning

3. Create a sprint plan up to the day of the final deadline (09.07.).
   → Using the Roadmap View in GitHub Projects
   → Example für Sprint durations, etc: Sprint Plan Template
   → Plan for the playtest on 12.06.!

Prepare for Production:
The Tools

Tools: Git & GitHub

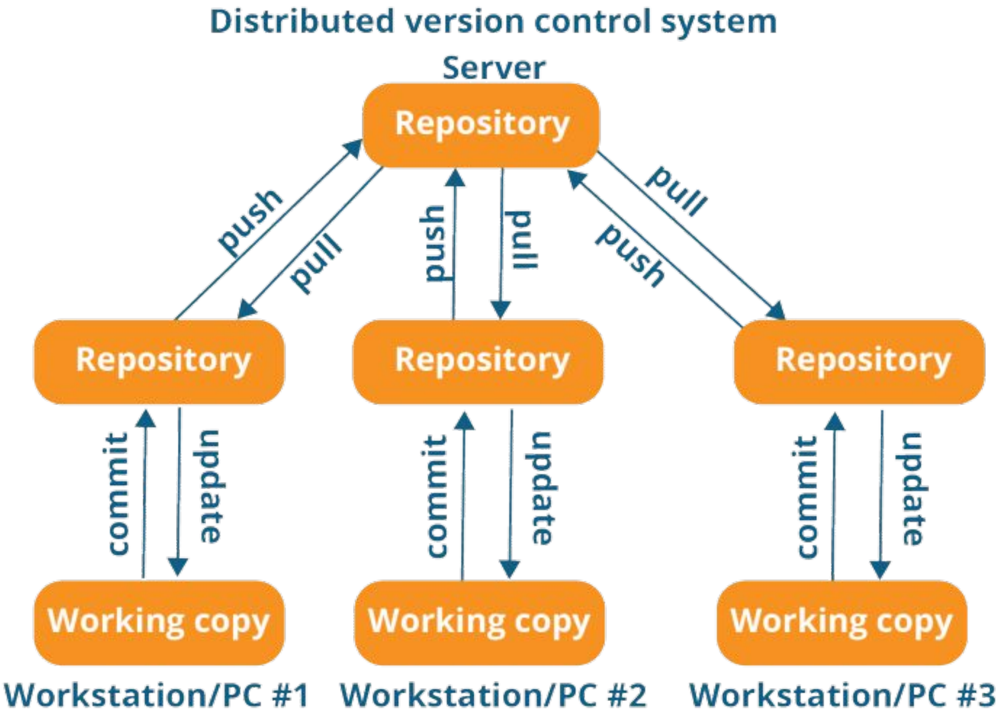
→ GitHub is where your code (Unity project) will live.
→ More on this in John's classes in April.
→ It is also the place where your project board exists.
→ You can use the Issues function of your repository to create stories that go into your project backlog.

Today, we'll create one repository per team/game and set up the project board.

Let's go!

Tools: Git & GitHub – What Is It?

→ Distributed Version Control
→ Your code is on a central server (GitHub).
→ Everyone in the team has a local copy of the repository.
→ You work on your local files and commit to your local repository.
→ At some point, you push it to the server for your team.
→ You also pull the changes that others made.

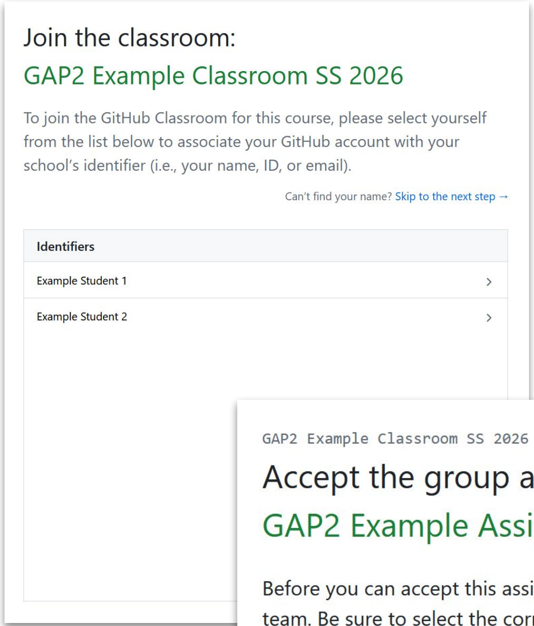


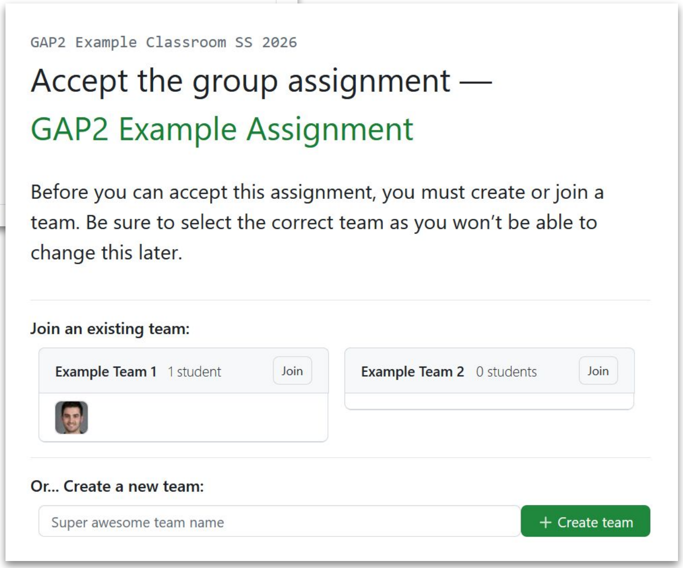
Tools: GitHub

- → Each team will receive their own repository on GitHub.
- → Workflow via **GitHub Classroom** (GHC)
  - → It’s a team assignment.
  - → Teams are already defined and can be joined.
  - → The whole team users a single repository.
  - → Each repository also has a project board connected to it.

Setting Up Your Repository

1. Make sure you're logged in to github.com
2. Accept the GitHub Classroom assignment (link in Moodle) and accept the GitHub permission request.
3. Choose your real name from the list.
4. Join the team that you assigned yourself to last week.
5. Follow the link to your repository.

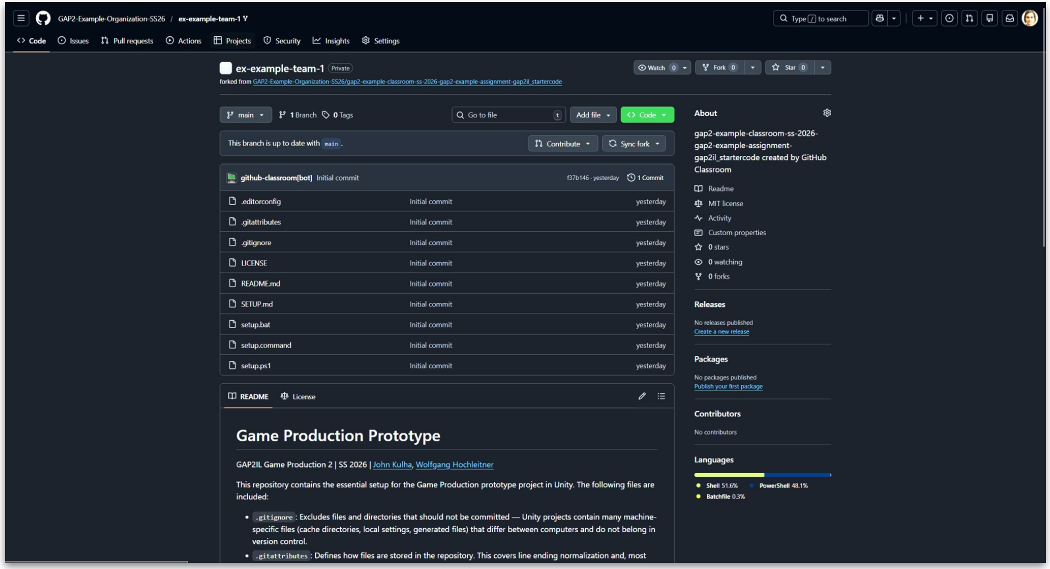




Your Repository

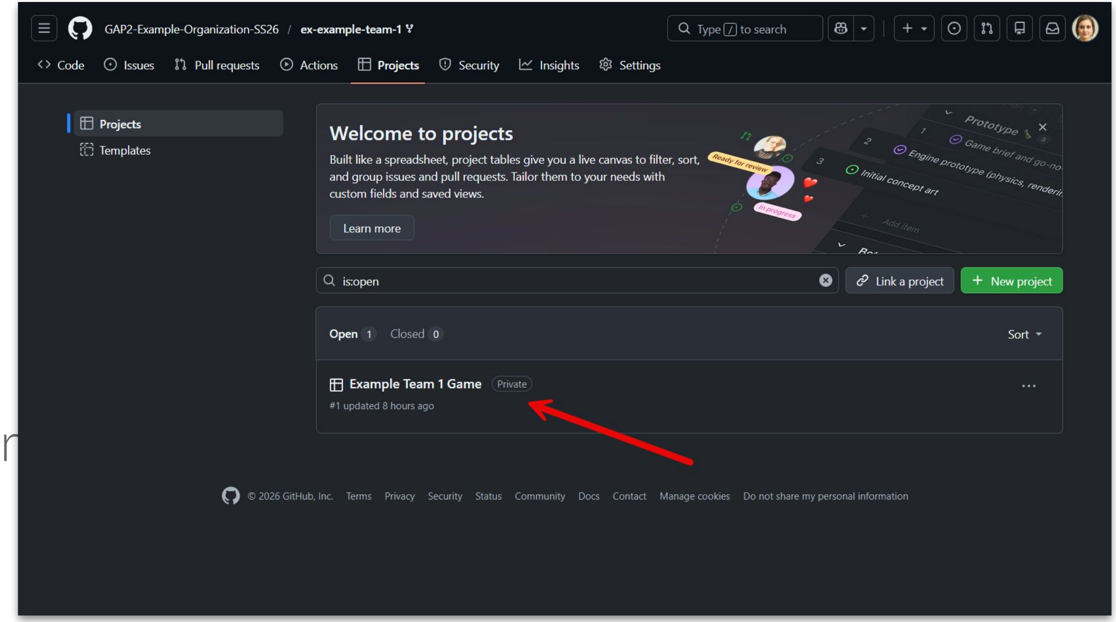
This is how your repository looks like.

1. Code shows the source code (files) of your project.
2. Issues can be used to track problems or things to do.
3. Projects is where your project board is.

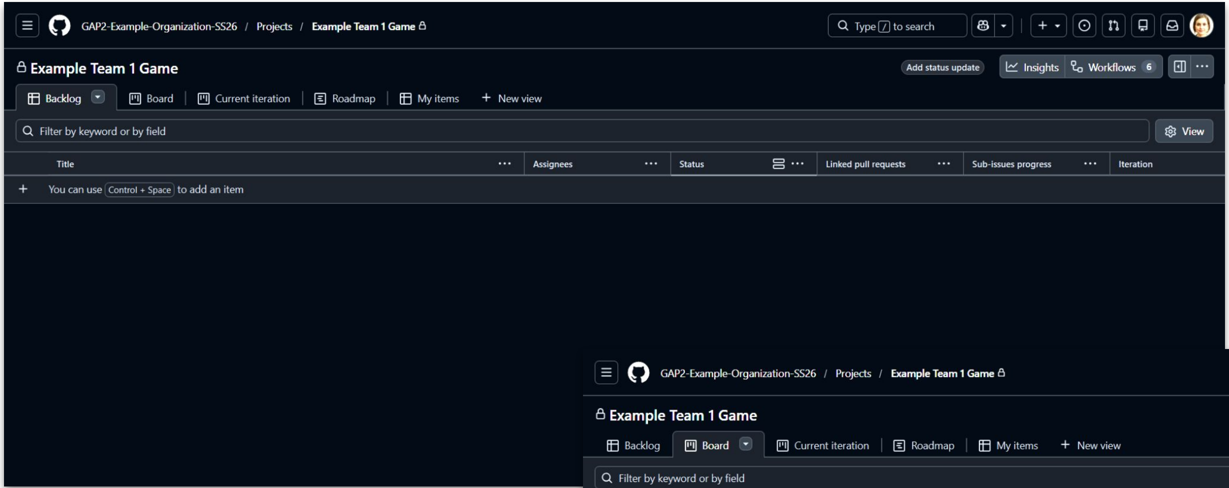


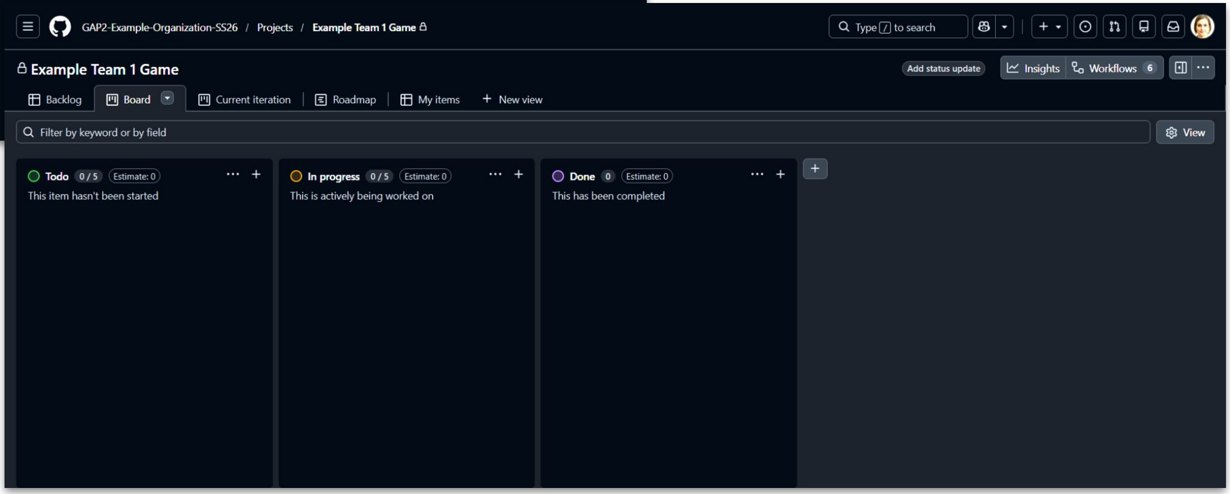
Your Project Board

1. Click on Projects in the upper menu bar of your repository.
2. Now choose the board that's listed here.
3. It's been set up so that your team has all the necessary permissions.



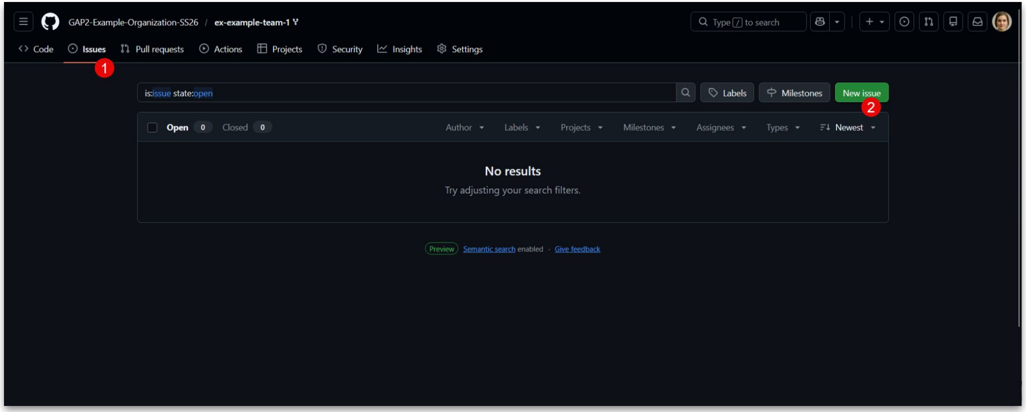
Your Board has different views.





Adding Your Stories and Tasks

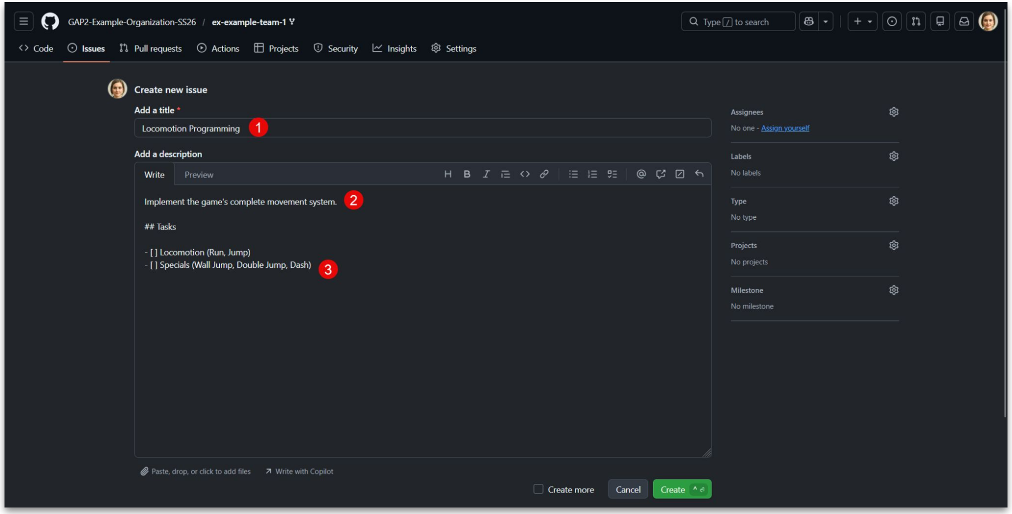
1. Every story is represented as an issue in GitHub.
2. Go to your repository, select the "Issues" tab and select "New issue" to create one.



Adding Your Stories and Tasks

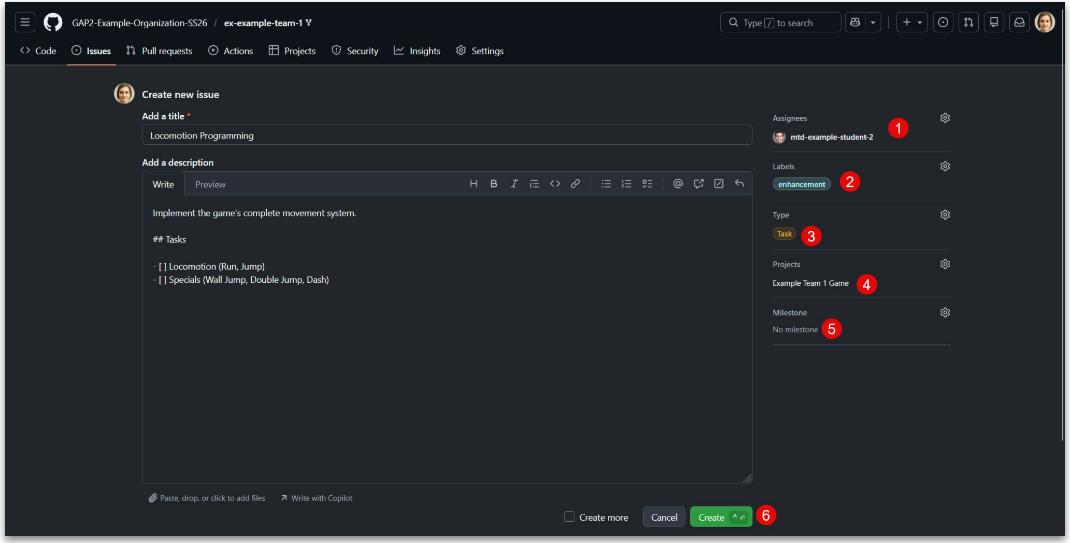
1. Choose a fitting name for your story.
2. Add a description to the field.
3. If you already know your tasks, add them as a tasklist by using a list in markdown combined with the square brackets: [ ]

```
- [ ] Task 1
- [ ] Task 2
```



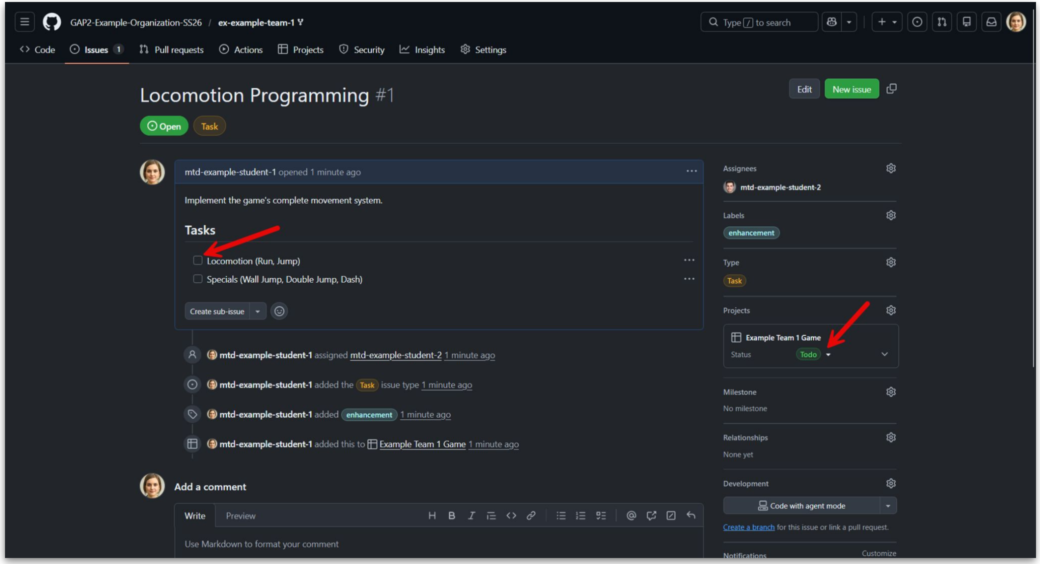
Adding Your Stories and Tasks

1. Assignee: Choose the person responsible for the story.
2. Labels: Optionally, label your story with attributes.
3. Type: Select the type of issue (e.g., feature or task)
4. Projects: Select your project to add it to the backlog.
5. Milestone: Optionally, create milestones. The act as a collection of issues/stories.
6. Create!



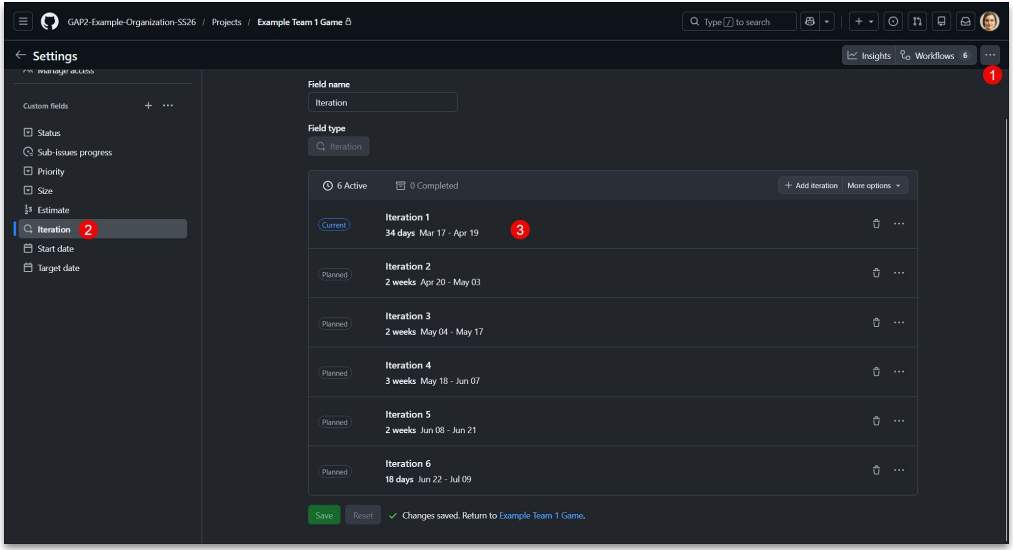
Adding Your Stories and Tasks

1. Have a look at your issue/story.
2. It can always be updated.
3. Tasks can be checked when done.
4. You can also convert them to a sub-issue (via "…" next to the task.)
5. The status in your project board is also updatable on the right.



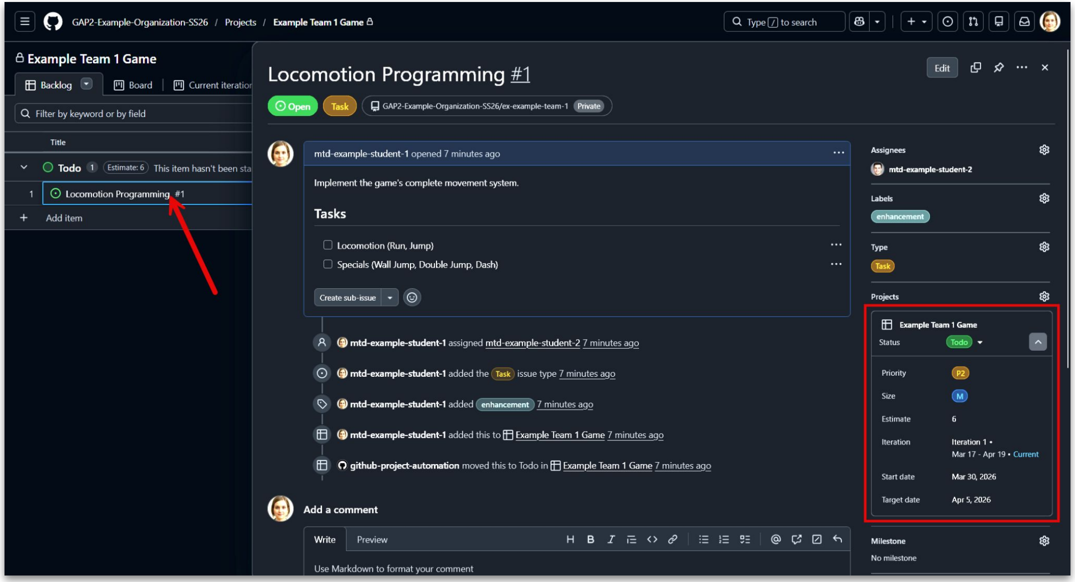
Configuring Sprints

1. Go back to your project and choose "... -> Settings"
2. Click on "Iteration" on the left (you can rename that to "Sprint" if you prefer that term).
3. Adapt the 3 predefined iterations and add more and save.



Set the Story Information

1. Select a story in your project board and open the details.
2. Choose Status (e.g., Todo).
3. Select Priority: P0 > P1 > P2
4. Select Size: XS, S, M, L, XL
5. Select Estimate: hours
6. Select Sprint & start and end date.



Use the tabs in your board to switch between views. Enjoy!

Tools: Communication

Setup infrastructure for online communication

→ Create a Discord server (or something similar) for your project team (if that hasn't happened already)

→ Hold one test-meeting with all team members

Use it from then on to communicate about your progress (standups, sprints).

Tools: Development Software

Install all the software so you can start to work on your game with John after the Easter holidays.

1. Unity 6000.3.11f
2. Git & Git LFS (Large File Storage)
3. GitKraken (or any other Git client you prefer)
4. P4Merge for resolving more stubborn merge conflicts.
5. Configure your repository for UnityYAMLMerge

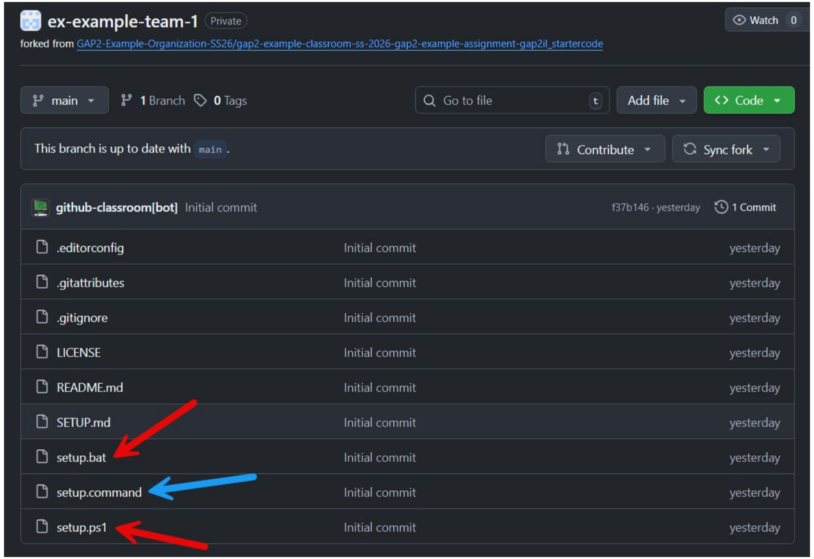
Tools: Development Software

Wait, what? That's complicated!

Unity: Install 6000.3.11 via Unity Hub

For the rest:

- → Windows: Download setup.bat and setup.ps1 and double-click setup.bat
- → macOS: Download setup.command and launch in Terminal
  - → `chmod +x setup.command` (to make it executable)
  - → `./setup.command` (to launch it)
  - → Double-clicking should work, but Apple's application protection will most likely not allow it.



This will install all the necessary software for you.

Clone your repository & configure Unity for Git

John will do this with you after Easter:
Once you've got all the tools installed, you can clone the repository to your disk.

Launch setup.bat / setup.command again, but this time from your repository again to configure the UnityYAMLMerge Tool for your repository (the script couldn't do that the first time).

Then follow this guide on how to set up a Unity project with git.
https://docs.google.com/document/d/11P1AcjXaPUuvcror2G9Sa_FPe-CttDB1ghrIzJWeXAw/edit?usp=sharing

To Dos

Project Management

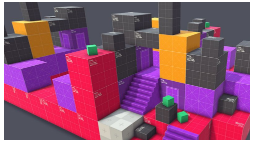
→ Access the repository & project board
  → Create stories (tasks, as you see fit)
  → Create a sprint plan (when is which story due?)
→ Set up communication
→ Install the software and clone your repository.

Concept & Presentation

→ Work on/finish your game concept
→ Present your concepts and project planning

Until Friday next week: 27.03.2026
(The tools can be installed until after Easter)

Thank you!



Title Graphic: Kenney Prototyping Textures
https://github.com/Calinou/kenney-prototype-textures